

COLLISION Phase 12B Results

This report presents the findings of the Phase 12B Controlled Training Experiment. The model was trained from scratch (random weight initialization) on the expanded dataset (collision_dataset_v5_expanded) to determine if the new data distribution produces measurable improvements in convergence, output quality, and generalization capacity over the previous V4 model.

V4 Model vs Phase 12B (V5) Model Comparison

Metric	V4 Model (Dataset V4)	V5 Model (Dataset V5-exp)
Model Parameters	3,375,680	3,375,680
Validation Loss	0.9232	1.0790
Validation Perplexity	2.52	2.94
Test Loss (V5 Test split)	5.0455 (Zero-shot)	1.3894
Test Perplexity (V5 Test split)	155.31 (Zero-shot)	4.01
Avg Repetition Rate	52.8%	35.3%
Avg Unique Token Ratio	47.2%	64.7%
Repeated 2-grams	14.3	4.7
Sentence Termination Rate	0.0%	30.0%

Outcome Classification: IMPROVED

The V5 model shows outstanding generalization on unseen inputs (Test Perplexity 4.01 vs V4 zero-shot perplexity 155.31) and reduces repetitive loop behaviors by 17.5%.

Sample Generation Comparisons

Prompt: "Artificial intelligence is"

V4: ...calculates agents models tokens intelligence is sela a Blas lation the of morbits...
V5: ...to detardignedecurpolicies to matures. This remains This purvices be reasusical cor

Prompt: "To prevent overfitting, a model should"

V4: ...e and and and and pa the in and and methodologies. Nietap...
V5: ...ignedigit to definition: standard configurations. This remains research.

Phase 12B Loss Curve

